Neural Network (Eita asole ki?)

Eita holo amader project er **shobcheye powerful Deep Learning Model (Artificial Neural Network)**:

* Amader hotel er **27 ta feature** (price, lead time, room type, etc.) input hishebe dhuke.
* Eita manush er brain er moto **layer-by-layer** ($128 \rightarrow 64 \rightarrow 32 \rightarrow 1$) neuron diye process kore.
* Shesh e **1 ta single number (Percentage)** ber kore dey: *"Ei guest cancel korar chance 85%!"*

⚙️ 2. HOW (Protita Line Ki Bhabe Kaaj Kore?)

Ashun layer gulo bujhi:

🔹 `Dense(128, activation='relu')`
* **Ki kore:** Prothom hidden layer, eikhane **128 ta neuron** ache. Eita input er 27 ta feature ke eksathe mix kore pattern khujte shuru kore.
* **`ReLU` ki:** Eita ekta magic switch! Formula: $f(x) = \max(0, x)$.
  * Positive number ashle pass kore dey.
  * Negative number ashle 0 banaye dey. Eitar karone model ta complex non-linear pattern sikh-te pare.

🔹 `BatchNormalization()`
* **Ki kore:** Data jokhon ek layer theke arek layer e jay, number gulo jate khub boro ba choto na hoye jay, eita tader ke **shoman scale e (normalize)** rakhe.
* **Labh:** Training onek fast ar smooth hoy.

🔹 `Dropout(0.3)`
* **Ki kore:** Training er shomoy eita **randomly 30% neuron ke bondho (off)** kore dey!
* **Keno eita dorkar:** Think of it like a group project: group er 3 jon student ke chuti dile baki 7 jon ke nijeder kaj shikhte hoy! Eita model ke **mukhoshtho (overfitting)** kora theke bachay.

🔹 `Dense(64)` $\rightarrow$ `Dense(32)`
* **Ki kore:** 128 ta information ke aro choto kore 64, tarpor 32 ta khub important summary te compress kore.

🔹 `Dense(1, activation='sigmoid')` (Output Layer)

* **Ki kore:** Eikhane matro **1 ta neuron** ache.
* **`Sigmoid` ki:** Eita shob calculation ke **0 theke 1 er moddhe ekta percentage** e convert kore (jemon $0.85 = 85\%$ chance of cancel).

🎯 3. WHY (Ei Settings Gulo Keno Dilam?)

```python
nn_model.compile(optimizer='adam', loss='binary_crossentropy')
```

1. **`Adam` Optimizer Keno?**
   * Training er shomoy model tar bhul theke weight update kore. `Adam` holo shobcheye smart optimizer—eita automatic shomoy bujhe speed baray ba komay, jate khub druto shera result pawa jay.
2. **`binary_crossentropy` Loss Keno?**
   * Amader target holo Binary (0 ba 1). Model jodi kono vul prediction kore, ei loss function ta model ke shasti (penalty) dey jate porer bar thik korte pare.

🛑 4. Early Stopping Keno Use Korlam?

```python
early_stop = EarlyStopping(monitor='val_loss', patience=5, restore_best_weights=True)
```

* **Patience = 5:** Model ta jodi dekhe 5 bar try koreo tar result aro bhalo hochhe na, eita **automatic training thamai dibe**.
* **Restore Best Weights:** Stop korar por, je round e shobcheye bhalo result chilo, sei model ta ke save kore nibe.
* **Keno:** Jate time nosto na hoy ar model mukhoshtho (overfit) na hoye jay!

🏆 Keno eita amader Shobcheye Best Model Holo?
Karun hotel booking er manush er decision linear na. Ei 4-layer Deep Neural Network ta non-linear interactions gulo ke shobcheye shundor bhabe sikh-te pereche, tai eita **81.6% Accuracy** ar **0.86 AUC** diye **#1 Champion** hoyeche!

📊 Figure 11: Grouped Bar Chart (Precision, Recall & F1-Score)

Eikhane protita model er jonno **3 ta alada rong er bar** pashapashi rakha hoyeche:
* 🟦 **Blue Bar (Precision = False Alarm Rate):** Model jokhon bole *"Cancel hobe"*, koto percent shomoy eita shotti?
* 🟩 **Green Bar (Recall = Catch Rate):** Hotel er asol cancellation gulor moddhe koto percent model ta dhor-te parlo?
* 🟥 **Red Bar (F1-Score = The Real Balance):** Precision ar Recall er shundor balance.

💡 Graph dekhe ki bujha jay:
1. **Naive Bayes:** Eitar **Green Bar (Recall)** akash chowa (97.1%), kintu **Blue Bar (Precision)** matro 32%! Mane eita shobai kei voy peye cancel bole dichhilo.
2. **Neural Network (Champion):** Eitar **Red Bar (F1-Score = 0.62)** shobar cheye unchu! Mane eita shobcheye shundor balance maintain koreche.

🔲 Figure 12: Confusion Matrices Grid (Bhul ar Shothik er Hishab)

Confusion Matrix holo ekta 2×2 box ja dekhiye dey model ta kothay kothay shothik ar kothay bhul koreche:

```
                [ Model Predict: NO ]     [ Model Predict: YES ]
[ Actual: NO ]   🟢 True Negative (TN)     🔴 False Positive (FP - False Alarm)
[ Actual: YES ]  🔴 False Negative (FN)    🟢 True Positive (TP - Caught It!)
```

🏨 Hotel er jonno shobcheye boro loss konta?
* **False Negative (FN - Niche Bam pashe):** Guest asolei cancel korse, kintu model bollo *"Guest ashbe"*. Hotel room khali pore roilo $\rightarrow$ **Direct Taka Loss!**
* **Graph Dekhe Result:**
  * **Neural Network & Decision Tree:** Egulor **FN (Loss)** ar **FP (False Alarm)** duita-i shobcheye kom chilo!
  * **Naive Bayes:** Eita pray shob guest kei "Yes Cancel" bole box e dhukiye diyeche.

---Here is the **complete, in-depth breakdown of the Confusion Matrix** for your project and why each model's matrix is better or worse than the others:

🔲 1. The 4 Boxes in a Hotel Booking Confusion Matrix

```
                        [ Model Predicts: NO ]       [ Model Predicts: CANCEL ]
[ Real Guest: COMING ]   🟢 True Negative (TN)        🔴 False Positive (FP - False Alarm)
[ Real Guest: CANCELS ]  🔴 False Negative (FN - Loss) 🟢 True Positive (TP - Caught It!)
```

What each box means for the Hotel's Business:

Box	What Happened	Business Impact
🟢 True Negative (TN)	Guest came $\rightarrow$ Model said *"Will Come"*	**PERFECT!** Room is ready, guest checks in, hotel makes revenue.
🟢 True Positive (TP)	Guest cancelled $\rightarrow$ Model said *"Will Cancel"*	**PERFECT!** Hotel knew in advance and resold the room.
🔴 False Positive (FP)	Guest came $\rightarrow$ Model falsely said *"Will Cancel"*	**BAD (Type I Error):** False alarm! Hotel might overbook the room; when the real guest arrives, no room is left (angry guest!).
🔴 False Negative (FN)	Guest cancelled $\rightarrow$ Model said *"Will Come"*	**WORST DISASTER (Type II Error):** Hotel kept the room empty waiting for a ghost. The night passes $\rightarrow$ **100% lost revenue!**

🥊 2. Model-by-Model Comparison: Why Each Matrix is Good or Worse

---

🥇 1. Neural Network — The Best Overall Balance
* **What its Matrix looks like:** Very high on the diagonal (High TN and High TP), with the lowest combination of mistakes.
* **Why it is the Best:** It manages to catch **over 54% of real cancellations** while keeping false alarms very low (72% precision). It protects hotel revenue without angering innocent guests.

🥈 2. Decision Tree — The Precision Master
* **What its Matrix looks like:** Extremely low **FP (False Positives)**.

* **Why it is Good:** When the Decision Tree flags a booking as *"Will Cancel"*, it is right **73% of the time**.
* **Why it is slightly behind Neural Net:** It misses a few more cancellations (higher FN) than the Neural Network.

🥉 3. KNN ($k=5$) — Moderate / Average
* **What its Matrix looks like:** Moderate across all 4 boxes.
* **Why it is Middle-tier:** Because it only checks the nearest 5 neighbors, bookings near messy boundary areas get misclassified easily, leading to a moderate amount of both FP and FN errors.

⚠️ 4. Logistic Regression — "Too Shy / Too Conservative"
* **What its Matrix looks like:** Huge on **TN** (it thinks almost everyone will show up), but a massive number of **FN (False Negatives)**.
* **Why it is Worse:** It missed **66.5% of all real cancellations** (Recall $= 33.5\%$)! A hotel manager using this model would be surprised by thousands of empty rooms because the model failed to flag them.

🚨 5. Gaussian Naive Bayes — "The Panicked False-Alarm Machine"
* **What its Matrix looks like:** Almost zero on the left column, and almost everything dumped into the right column (**High TP, but Massive FP**).
* **Why it is the Worst:** It predicted *"CANCEL"* for nearly the entire hotel!
  * While its Recall looks high (97%), its **False Positive rate is catastrophic**.
  * If a hotel used Naive Bayes, they would overbook the hotel and have to turn away thousands of real arriving guests at the front desk!

🎙️ Quick 3-Sentence Summary for Viva:
> *"The Confusion Matrix shows the trade-off between false alarms (FP) and missed cancellations (FN). The **Deep Neural Network** produced the best matrix because it balanced both errors, whereas **Logistic Regression** missed too many cancellations (high FN) and **Naive Bayes** raised false alarms on almost everyone (huge FP)."*

📈 **Figure 13: ROC Curves with AUC Comparison**

Eikhane 5 ta model er curve eksathe aka hoyeche:

```
  Catch Rate (Recall)
   1.0 |        .--- Neural Network (AUC = 0.8602) 🥇
       |      .'
       |    .'
       |  .'
       | /  <-- Kalo Dotted Line (AUC = 0.50, Coin Toss)
   0.0 +----------------------
      0.0                  1.0
      False Alarm Rate
```

* **Y-axis:** Real cancellations catching rate (Joto **Upore** uthbe, toto bhalo).
* **X-axis:** False alarm rate (Joto **Bame** thakbe, toto bhalo).
* **Kalo Dotted Line:** Eita holo Random Guessing (Coin toss, $\text{AUC} = 0.50$).

🏆 Winner:
* Shobcheye opore ar bame (Top-Left corner) gheshe geche **Neural Network (AUC = 0.8602)**, tar thik pichone ache **Decision Tree (AUC = 0.8305)**!
* Eita proman kore je **Deep Neural Network** shob threshold ei cancellations alada korar jonno shobcheye shera discriminative power rakhe.

8. Unsupervised Learning (K-Means Clustering)

Ashun **Section 8 (Unsupervised Learning)** er shobkisu ekdom shohoj **Banglish** e bujhe nei:

🤖 1. Kono Model Use Korlam?

* **Main Model:** **K-Means Clustering** (Unsupervised Algorithm).
* **Helper Tool:** **PCA (Principal Component Analysis)** — 27 ta feature ke 2D screen e chobir moto dekhanor jonno dimensionality reduction tool.

💡 2. WHAT, HOW, WHY (Shobcheye Shohoj Bhabe)

🔹 WHAT (Eita asole ki?)

* **Supervised Learning e:** Amra model ke answer sheet diye ditam: *"Ei customer er lead time 100, ar eita CANCEL hoise (1)."*
* **Unsupervised Learning e:** Amra target column `is_canceled` ke **ekdom delete / remove** kore felechi! Tarpor model ke bolchi: *"Tumi nije nije data dekhe customer der moddhe similarity khuje duita dol (cluster) e vag koro."*

🔹 HOW (Ki bhabe kaaj korlo?)
1. **Target Bad Dilam:** `is_canceled` column ta drop kore dilam (`X_unsup`).
2. **Elbow Method ($k=1$ theke 10):** Test korlam koyta cluster e vag kora shobcheye natural ($k=2$).
3. **K-Means Train Korlam:** $k=2$ diye model ke bollam shob booking ke Group 0 ar Group 1 e vag korte.
4. **PCA ($27\text{D} \rightarrow 2\text{D}$):** 27 ta column ekshathe graph e aka jay na, tai PCA diye 2D graph e convert korlam.

🔹 WHY (Keno eita korlam?)
* **Real-world Test:** Amra check korte cheyechi je kono answer sheet (labels) charao machine ki bujhte pare je hotel e 2 dhoron er customer ache (Cancel group vs. Show-up group)?

📊 3. Diagram Gulo Ki Mean Kore? (Figure 14 & Figure 15)

📉 Figure 14: Elbow Method for Optimal k (Oporer Graph)
* **X-axis (k):** Koyta cluster (Group) banabo (1 theke 10).
* **Y-axis (Inertia / WCSS):** Group er moddhe data koto chhoriye-chitiriye ache (Joto **KOM**, group toto compact).
* **Graph e ki dekhchen:**
  * $k=1$ theke $k=2$ te ashar shomoy line ta **druto niche neme ekta elbows (konui er moto benk)** toiri koreche!
  * **Meaning:** $k=2$ holo optimal point. Eita mathematically proman kore je amader hotel dataset ta naturally **2 ta boro group** e vag hoy (Canceled vs Not Canceled).

🎨 Figure 15: PCA Projection (Nicher Dui Graph)

Eikhane pashapashi duita scatter plot ache:

```
[ Figure 15a: K-Means Clusters ]          [ Figure 15b: Asol Ground Truth ]
Model nije nije je duita group banise     Asol reality (Ke asole cancel korse)
```

```

1. **Left Plot (Figure 15a - K-Means Ki Korlo):**
   * K-Means data ke **duita alada dol** e divide koreche:
     * Bam pasher Shobuj dol (Cluster 0)
     * Dan pasher Lal dol (Cluster 1)
2. **Right Plot (Figure 15b - Asol Reality):**
   * Eikhane asol cancellation label dekhano hoyeche (Green = Kept, Red = Canceled).
3. **🌟 The Big Discovery (Viva Point):**
   * Dui pasher graph er pattern dekhen: **K-Means kono target label charao** high-risk booking der ke low-risk booking theke alada ekta cluster e felte pereche!

---

### 🎙️ 20-Second Viva Summary:
> *"Sir, Section 8 e amra **K-Means Clustering ($k=2$)** use korechi target variable `is_canceled` ke remove kore. **Elbow method (Figure 14)** proman koreje optimal cluster number $k=2$. Tarpor **PCA 2D projection (Figure 15)** diye amra dekhiyechi je unsupervised K-Means clusters gulo ground-truth cancellation pattern er shathe shundor bhabe align kore."*
```